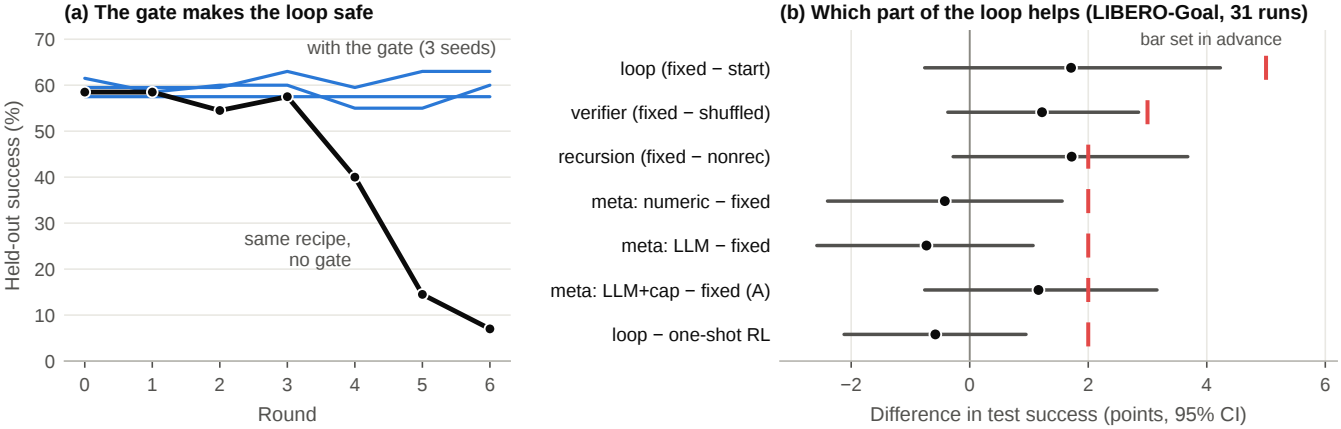
(a) The gate makes the loop safe
with the gate (3 seeds)
same recipe,
no gate
Held-out success (%)
Round
0
10
20
30
40
50
60
70
0
1
2
3
4
5
6
(b) Which part of the loop helps (LIBERO-Goal, 31 runs)
bar set in advance
loop (fixed − start)
verifier (fixed − shuffled)
recursion (fixed − nonrec)
meta: numeric − fixed
meta: LLM − fixed
meta: LLM+cap − fixed (A)
loop − one-shot RL
−2
0
2
4
6
Difference in test success (points, 95% CI)